ECON 4660 / 8666 — Problem Set 1

International Economic Development · Fall 2026 · Prof. Zhigang Feng
Due Monday, September 21, before class (Canvas). At most 3 pages. Do not attach raw spreadsheets. State the data source and the exact series name for every number.

This set practices the data skills and ideas that will show up on the midterm. The midterm will use different countries and will not repeat the growth-accounting question.

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1. Living standards in the data (World Bank)

Use World Bank Open Data (no login). Countries: China, India, Korea, Rep., Thailand (not the midterm countries). Years: 1990 and 2019. If a cell is missing, use the nearest year and say so. Do not interpolate.

Series

1. GDP per capita, PPP (constant 2017 international $) — or the current PPP constant series if 2017 is retired; write the exact name.
2. Life expectancy at birth, total (years).
3. Fertility rate, total (births per woman).

Hints. Search the indicator name, then pick the four countries. You can also use WDI DataBank.

(a) One table: for each country, 1990 and 2019 values of all three series, and the average annual growth rate of GDP per capita

$$g = \left(\frac{y_{2019}}{y_{1990}}\right)^{1/29} - 1.$$

If your window is not 29 years, change the exponent.

(b) Rank by 1990 GDP per capita (poorest first) and by g. In this sample, do poorer countries grow faster? One short paragraph: what Solow theory says about *unconditional* vs. *conditional* convergence, and why four countries are not a test.

(c) Rank by the gain in life expectancy. Same ranking as income growth? One short paragraph on growth vs. development.

ECON 8666 only (½ page). Add poverty headcount at $3.00 a day (2021 PPP), WDI `SI.POV.DDAY` (or PIP). These are survey years, not annual. Use the nearest years to 1990 and 2019. Which country turned growth into poverty reduction most clearly? One channel from Ch. 6.

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2. Growth accounting, 1990–2020

Same four countries. This question will not be repeated on the midterm; the exam will ask you to *interpret* growth accounting, not to recompute it.

Use the Penn World Table (PWT) for real GDP, capital stock, and employment (WDI does not have a usable capital-stock series). Name the PWT variables you used (for example `rgdpo`, `rnna`, `emp`).

Let Y_t be real GDP, K_t capital, L_t employment. Assume $\alpha = 0.33$ in all countries.

(a) TFP: $A_t = Y_t/(K_t^{\alpha} L_t^{1-\alpha})$.

(b) Annual growth rates of Y, K, L, and A.

(c) Contribution of capital to GDP growth in a given year: $\alpha g_{K,t}/g_{Y,t}$ (and analogously for labor and TFP). You may report averages over 1990–2020 instead of every year.

(d) Interpret. Which margin—capital, labor, or TFP—did most of the work in each country? One paragraph.